

Symbolic and Numeric Differentiation

1 Symbolic and Numeric Differentiation

- symbolic differentiation
- numeric differentiation

2 Implicit Differentiation

- the tangent line at a point on a curve
- implicit differentiation to compute the slope

MCS 320 Lecture 22
Introduction to Symbolic Computation
Jan Verschelde, 13 July 2026

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Symbolic Differentiation

Example: compute the value of the derivative of $f = \sin(2\pi x)$ at $x = 1$.

This is a 2-step process:

$$\textcircled{1} \quad f'(x) = \frac{d}{dx} \sin(2\pi x) = \cos(2\pi x)2\pi,$$

$$\textcircled{2} \quad f'(1) = 2\pi.$$

Observe that the value in the example is a symbol.

We can describe this process in general as follows.

Given an expression f which depends on a variable x , the result of *symbolic differentiation* is an expression f' for the derivative of f with respect to x .

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Numeric Differentiation

Given a function $f(x)$, and a value A for x ,
the *central difference formula*

$$\text{for } h > 0 : \quad \frac{f(A + h) - f(A - h)}{2h} \approx f'(A)$$

approximates $f'(A)$, the value of the derivative of f at $x = A$.

The error of this approximation is $O(h^2)$.

Two concerns that are addressed by numerical analysis:

- The accuracy depends on a good choice for the step h , and the precision at which f is evaluated.
- Numerical extrapolation methods improve the accuracy.

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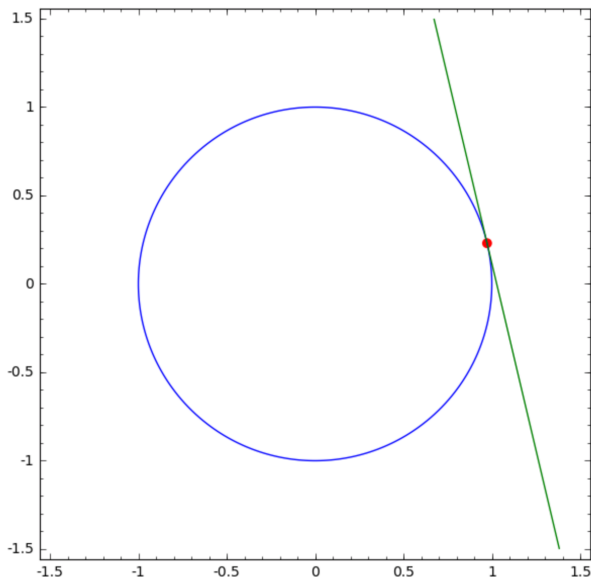
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The Tangent Line at a Point on a Curve



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Implicit Differentiation to Compute the Slope

Input: $f(x, y) = 0$, an equation and a point $(a, b) : f(a, b) = 0$.

Output: $\left. \frac{dy}{dx} \right|_{(x,y)=(a,b)}$ is the slope of the tangent line
to the curve $f(x, y) = 0$, at the point (a, b) .

The three steps to compute the slope of the tangent line are below.

- 1 View $y = y(x)$ as a function of x .
- 2 Apply the operator $\frac{d}{dx}$ to the equation $f(x, y(x)) = 0$:

$$\frac{df}{dx} + \frac{df}{dy} \frac{dy}{dx} = 0.$$

- 3 Solve the equation for $\frac{dy}{dx}$.